

Daniella Toth

Software Engineering Student Seeking Winter 2027 SWE Internships/Co-ops

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EDUCATION

University of Guelph

Guelph, ON

Honours Bachelor of Computing, Software Engineering (Co-op)

Sept. 2023 – April 2028

- **Relevant Coursework:** Software Engineering, Software Design, Database Systems & Concepts, Data Structures & Algorithms, Operating Systems, Software Reliability & Testing, Human-Computer Interaction, Mobile Computing

EXPERIENCE

Front-End Web Developer (Co-op)

August 2025 – December 2025

DBM Systems Inc.

Hybrid, Waterloo, ON

- Contributed to DBM's S&OP planning web app, used internally by 14+ enterprise clients, shipping 1 major page and 2 additional pages in Svelte, TypeScript, and CSS.
- Built a base table cell component and 3 cell types (numeric, string, percentage) with shared editing and validation logic, reused across the app.
- Implemented spreadsheet-level interactions: keyboard navigation, multi-cell selection, copy/paste, inline editing, and dynamic column resizing.

Undergraduate Teaching Assistant

Sept. 2024 – Apr. 2026

University of Guelph

Guelph, ON

CIS*2250: Software Design II

Jan. 2026 – Apr. 2026

- Supported students in a project-based software design course emphasizing collaboration and industry best practices.
- Ran labs and debugged Python-based team projects, helping students reason through design decisions and trade-offs.

CIS*1500: Introduction to Programming

Sept. 2024 – Apr. 2025

- Supported 100+ students through weekly labs, office hours, and discussion forums, debugging code and explaining core programming fundamentals.

PROJECTS

Kerr Black Hole Simulation | Python, OpenGL/GLSL, GLFW, NumPy

May 2026 – Sept. 2026

- Co-developed a real-time GPU-shader renderer, integrating null geodesics via RK4 (up to 2,500 steps/pixel) in Kerr-Schild coordinates at 720x480, with ImGui controls for live disc temperature and spin.
- Computed disc temperature across 1,024 radial points via a relativistic flux model, mapped to colour using Planck's law, CIE colour-matching, and gravitational redshift, rendered against a 119,625-star HYG starfield.

TuneTwin | Python, Flask, Last.fm API, Deezer API

Nov. 2025 – Present

- Co-developed and deployed a full-stack Flask music recommendation app, calling the Last.fm API for similar-track discovery and the Deezer API for album artwork.
- Returns up to 10 matches per search in an interactive swipeable carousel, plus a homepage 'tonight's pick' feature that reshuffles without a page reload.

ElementLab | Vue.js, FastAPI, PostgreSQL, scikit-learn

Jan. 2026 – May 2026

- Built a full-stack weather analysis platform as Data Engineer & Frontend Developer on an 8-person team, querying 6.4M rows of historical weather data.
- Trained a k-nearest-neighbours model reaching an R^2 of 0.756 for temperature and up to 0.819 for precipitation prediction.

EXTRACURRICULARS

Guelph Student Investment Council (GSIC)

Sept. 2025 – Apr. 2026

Analyst

- Analyze public companies and evaluate financial statements, business models, and industry trends to prepare investment theses for council review.

TECHNICAL SKILLS

Languages: C, Python, Java, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Libraries: Svelte, Vue.js, Flask, FastAPI, OpenGL, GLFW, ImGui

Data & ML: PostgreSQL, scikit-learn, k-nearest neighbours

Tools & Practices: Git, GitHub, VS Code, Docker, REST APIs, Agile Development